

Network Traffic Analysis

Suspicious Traffic Identification Report

Interface: NAT (Eth1) | Tools: Wireshark & tcpdump

Introduction

During this analysis, Wireshark and tcpdump were used to capture HTTP, DNS, and ARP traffic on the NAT interface (Eth1). Traffic was generated using curl requests to a known vulnerable test site, and filters were applied to isolate each protocol for detailed inspection.

HTTP Traffic Observations

- A curl request was made to `http://testphp.vulnweb.com`, a known vulnerable test site.
- The HTTP GET packet revealed: request method GET, Host header `testphp.vulnweb.com`, User-Agent `curl/x.x`, and a response code (e.g. 200 OK or 404).
- Suspicious: Unencrypted HTTP traffic (no HTTPS) means data is sent in plaintext — credentials or sensitive data could be intercepted by anyone on the network.

DNS Traffic Observations

- DNS queries were observed resolving domain names to IP addresses over the NAT interface.
- Standard A record lookups were seen for `testphp.vulnweb.com`.
- Suspicious: Any DNS queries to unknown or unusual external domains could indicate DNS tunneling or C2 (command-and-control) communication.

ARP Traffic Observations

- ARP Requests were broadcast across the network asking 'Who has IP x.x.x.x?'
- A corresponding ARP Reply confirmed the MAC address of the target host.
- Suspicious: Excessive or repeated ARP requests from a single source could indicate ARP scanning or ARP poisoning/spoofing attempts.

Conclusion

The most notable concern identified during this capture is the use of unencrypted HTTP traffic. In a real-world network environment, this would expose sensitive data to potential

interception. ARP and DNS traffic appeared mostly normal during the capture period, but both protocols should be continuously monitored for anomalies such as unexpected domain resolutions or ARP flooding, which are common indicators of network attacks.